SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5_AT2 — VISUALIZATION ANALYSIS REPORT

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO5 – Evaluation (BL5)	**Assessment:**	Assessment Tool 2 – Visualization Analysis Report
Weightage:	20% of CIA	**Total Marks:**	25 (5 Questions × 5 Marks)
CO5 Statement:	*Evaluate the effectiveness of visualization design choices — proportional ink, axis scales, overplotting solutions, color use, and accessibility — and justify improvements.(BL5)*		
Student Name:	Nivya Sri T	**Reg. No.:**	192424131
Section / Batch:	2024	**Date:**	27-08-26

Code of Conduct

I ______Nivya Sri T_(192424131)__________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _________*Nivya Sri T*_________________

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: explain WHY the design choice is flawed, and justify your judgment.
- Where a question asks for a fix, propose a specific improvement — not just “make it better.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION ANALYSIS REPORT

Rubrics for Calculation **ASSESSMENT RUBRIC**

AT2 — Visualization Analysis Report

CO5 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO5-AT2 Visualization Analysis Report problems, in which students critique a described visualization design choice, identify its specific flaw, and justify a well-reasoned improvement.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	**25%**	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., proportional ink, axis scale choice, overplotting, color scale design, accessibility).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	**25%**	Clearly explains why the design choice is flawed or effective, including its practical consequence for a reader — not just that it is ‘good’ or ‘bad.’	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	**20%**	Proposes a specific, well-justified fix or recommendation that directly addresses the evaluated issue.	Proposes a workable fix, with only minor gaps in justification.	Proposes a fix, but it only partially addresses the issue or lacks justification.	Fails to propose a workable fix, or it does not address the evaluated issue.
Organization	**15%**	The response is clearly structured, addressing critique and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	**10%**	Both the critique and the recommendation	Mostly accurate, with only minor omissions or a	Partially correct; the critique or recommendation	Largely incorrect or inappropriate, with major gaps

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		are completely accurate and well-suited to the specific scenario described.	small error.	contains notable inaccuracies.	in both the critique and the recommendation.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

PROBLEM 36: Contour Lines

What tightly-packed contour lines indicate

- When contour lines are close together, the density is changing rapidly.
- In this context, it means:
  - A steep gradient in the relationship between study hours and exam scores.
  - Scores increase sharply with small increases in study hours.
  - The region has high concentration of students with similar study hours and scores.

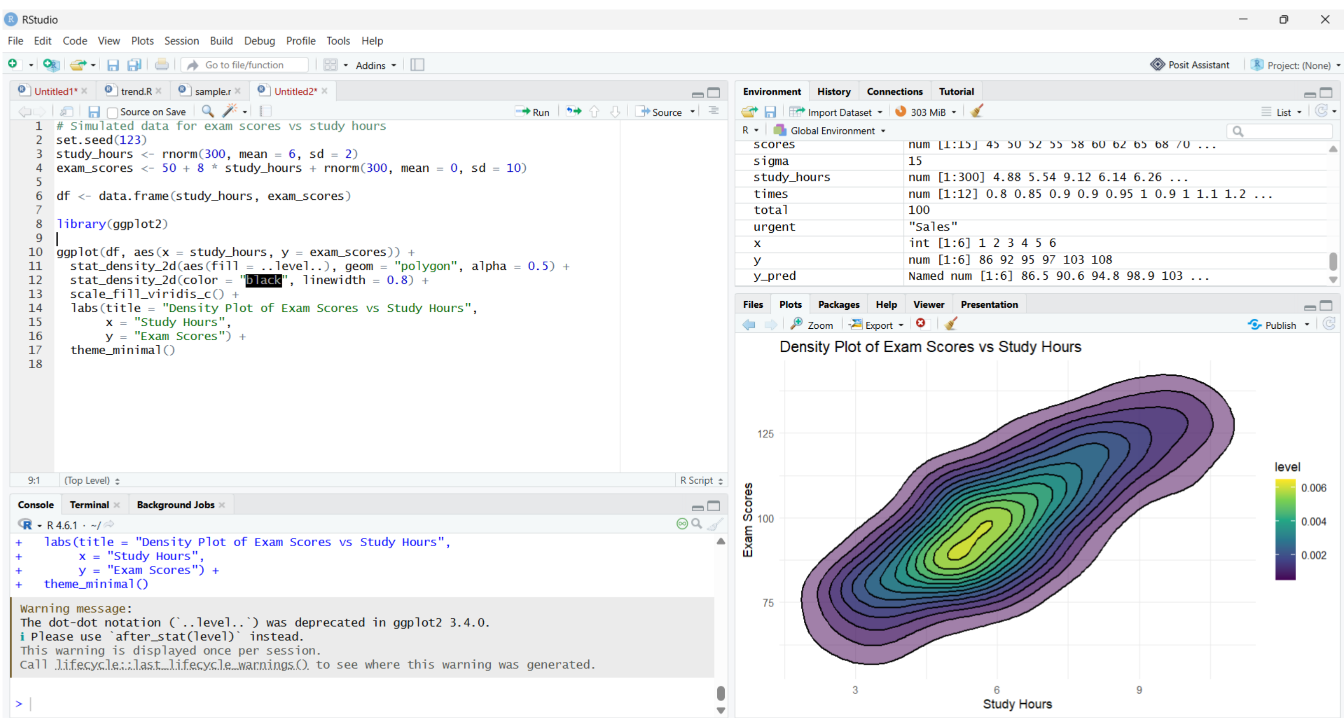
What widely-spaced contour lines indicate

- When contour lines are far apart, density changes slowly.
- This means:
  - A gentler relationship between study hours and exam scores.
  - Scores vary more gradually as study hours change.
  - The region has lower concentration of students.

✓ Final Interpretation

- Tightly-packed lines = strong, rapid change → steep relationship
- Widely-spaced lines = slow change → weaker or flatter relationship

This helps viewers understand how exam performance responds to study time, beyond what a simple scatter plot would show.



PROBLEM 37: Density Estimation Techniques

Advantage of KDE over histogram

A KDE (Kernel Density Estimate) provides a smooth, continuous estimate of the distribution. This has one major advantage:

A KDE does not depend on arbitrary bin choices, whereas a histogram's shape can change dramatically depending on the number and width of bins.

So the KDE often gives a clearer, more stable picture of the true underlying distribution.

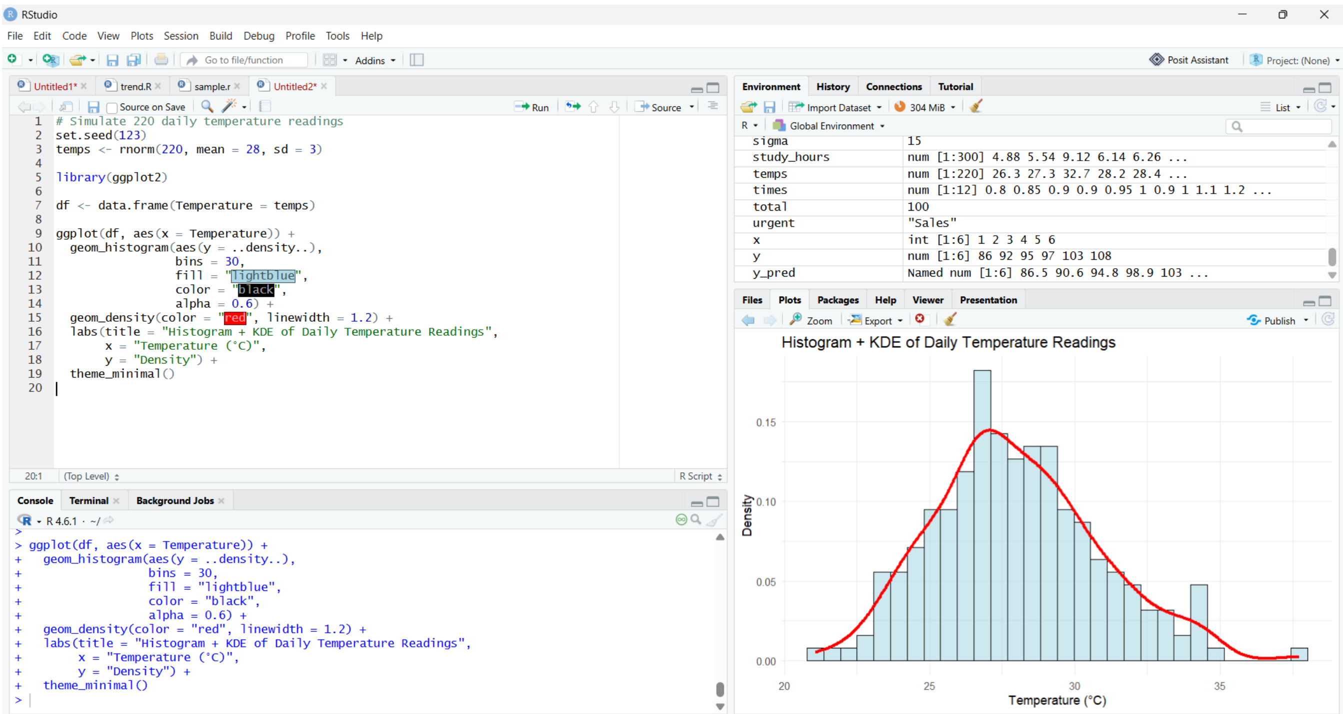
What happens if the KDE bandwidth is set too wide?

If the bandwidth is too wide:

- The KDE becomes over-smoothed
- Peaks and valleys disappear
- Important features (like multimodality or skewness) get flattened
- The curve may look almost like a straight line

In short:
Too wide a bandwidth hides meaningful structure in the data.
✓ Final Interpretation

- KDE advantage: smooth, bin-free, stable distribution estimate
- Too-wide bandwidth: oversmoothing → loss of detail



PROBLEM 38: Misuse of Color

Critique of inconsistent colors
When two related bar charts use different colors for the same response categories, viewers get confused because:

- They cannot track categories across charts
- Visual memory breaks — "Agree" might be green in one chart and blue in another
- Interpretation becomes slower and error-prone
- The chart violates the principle of color consistency, a core rule in data visualization

In short:
Inconsistent colors destroy comparability between charts.
Purposeful, consistent alternative
A better approach is to use a systematic palette where colors encode meaning:

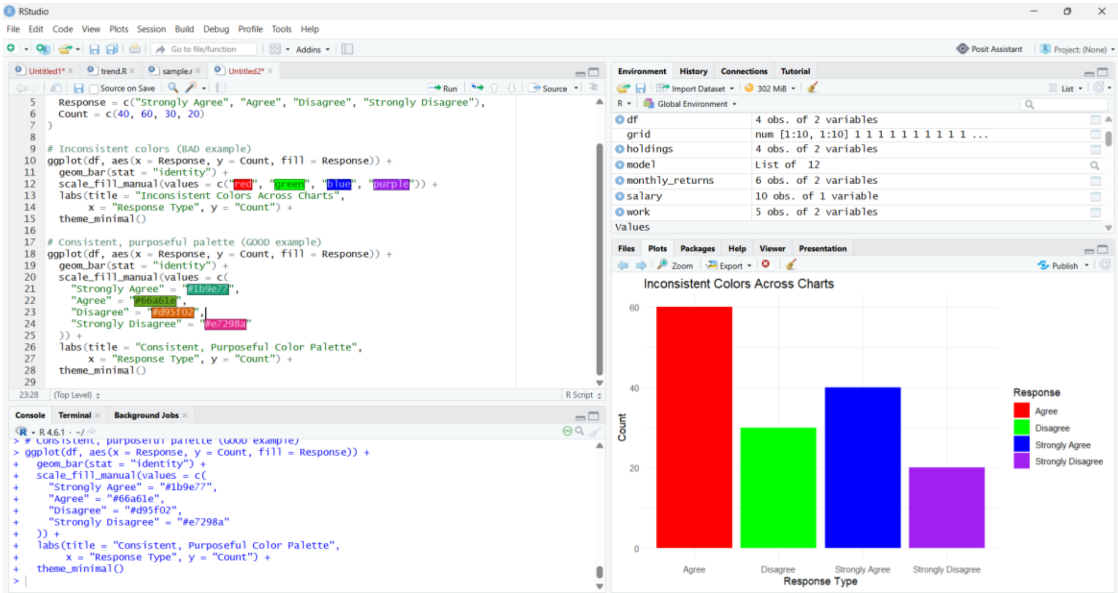
- Greens → positive responses (Strongly Agree, Agree)
- Oranges/Pinks → negative responses (Disagree, Strongly Disagree)

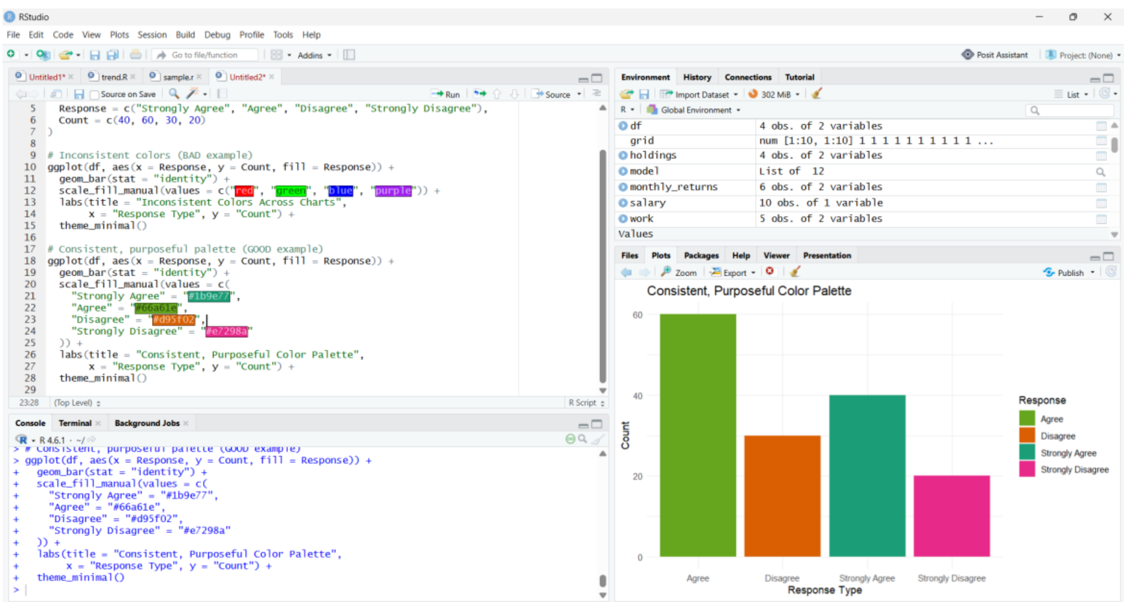
This way:

- The same category always has the same color
- The palette communicates semantic meaning
- Viewers instantly recognize patterns across multiple charts
- Comparisons become faster and more intuitive

✓ Final Interpretation

- Bad choice: Random colors → confusion, no comparability
- Good choice: Consistent, meaningful palette → clarity, faster interpretation





PROBLEM 39: Encoding Too Much or Irrelevant Information

1. Is the 5-encoding, heavily annotated design effective? For a general audience, no. Five simultaneous encodings (position, color, shape, size, transparency) plus overlapping annotations:

- Overload working memory
- Make it hard to see the main pattern
- Cause visual clutter and label collisions
- Force viewers to decode a legend instead of seeing the story

So the design is too complex for non-experts.

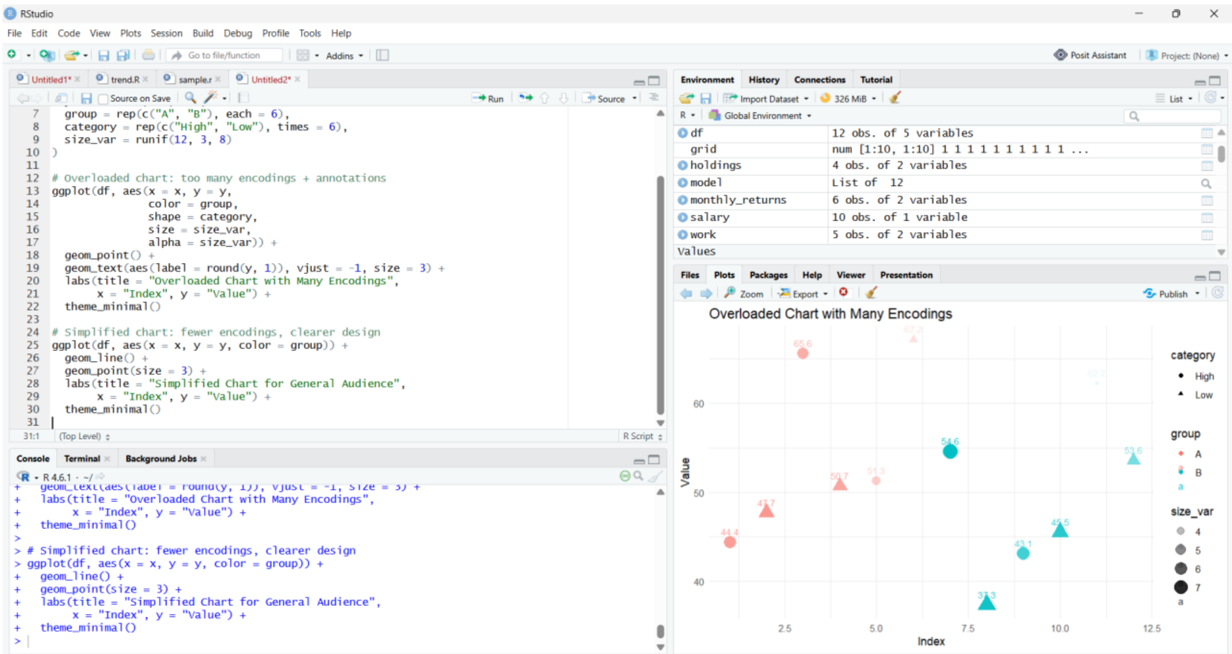
2. Proposed simplification

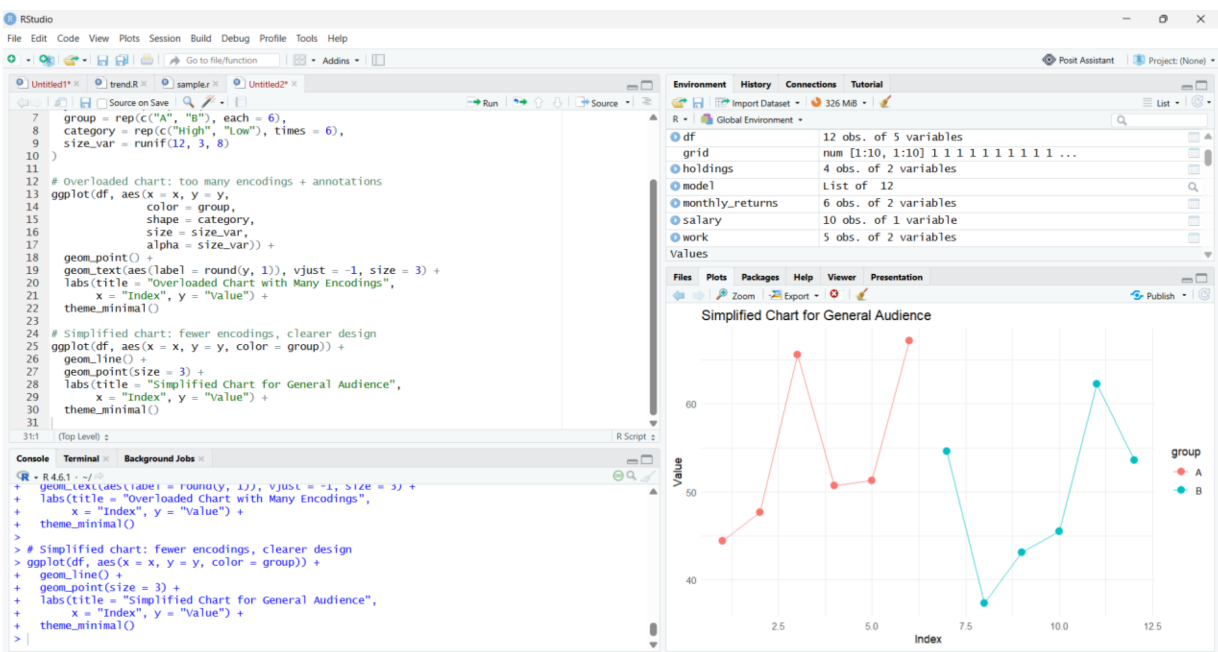
A clearer design would:

- Use 2–3 encodings only (e.g., position + color, maybe line vs points)
- Remove most text labels, keep only key annotations or a legend

- Focus on the main comparison (e.g., two groups over 12 points)
- Optionally split extra variables into separate small multiples (faceted charts)

This way, the chart becomes readable, focused, and suitable for a general audience, while still conveying the essential message.





PROBLEM 40: Avoiding Nonmonotonic Color Scales

Critique of inconsistent colors

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